Fuel Composition as a Dynamic Control Variable in Spark-Ignition Engines: Chemistry, Operating Regimes, and the Limits of Blend Scheduling

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Abstract

This paper formalizes and tests a simple thesis: *fuel chemistry is an engine control variable.* Different fuel molecules do not merely shift the timing of heat release; they change its shape (the rate of pressure rise, the location of peak pressure, the flame temperature) and therefore change which operating regimes an engine can enter. Using zero-dimensional chemical kinetics (Cantera) coupled to a calibrated knock-limited spark-ignition engine model, the study asks where, if anywhere, *dynamic* fuel-composition scheduling buys efficiency: across operating points, across ambient conditions, and across equivalence ratio into lean operation. Six fuels and blends (gasoline surrogate, ethanol, hydrogen, methane, ammonia, ammonia/hydrogen) are evaluated with validated mechanisms (LLNL gasoline, GRI-Mech 3.0, CRECK ammonia). The results are threefold. First, at stoichiometric operation, per-operating-point blend scheduling is worthless (+0.015 efficiency points (Modelled, B), indistinguishable from the screening model's numerical noise and read as zero): the optimum is a corner solution, and the prize is fuel choice plus compression ratio co-optimized once (+1.20 points vs. a realistic gasoline engine (Modelled, B); the second decimal is indicative). Second, ambient-adaptive fuel selection is worth +0.05 points (Modelled, B), a margin within the mechanism cross-comparison uncertainty and likewise read as zero: the winning fuel does switch with ambient, confirming the mechanism, but the top fuels sit within half a point everywhere. Third, in lean operation the answer flips: hydrogen's unique ultra-lean capability opens a high-γ, low-pumping cruise regime that misfire-walled hydrocarbons cannot enter, and load-adaptive switching between hydrogen and a hydrogen/methane blend is worth +0.95 points (+1.44 drive-weighted) over the best fixed fuel (Modelled, B; the adverse sensitivity variant bounds the range at +0.44/+0.75 points, so the sign is firm and the magnitude lies between roughly one-half and one-and-one-half points). The refined thesis: chemistry's value is not in reshaping the burn at a fixed point but in *reaching operating regimes.*

Keywords: dual-fuel engines, fuel blending, hydrogen, lean combustion, knock, engine efficiency, chemical kinetics

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1 Introduction

The control variables of a conventional engine are air, fuel quantity, spark timing, and valve events. Fuel *identity* is treated as a fixed property of the machine: an engine is built for gasoline, or diesel, or natural gas, and the chemistry is then taken for granted. This paper starts from the observation that this need not be so. Different chemical reactions create different types of pressure and different types of heat. Hydrogen releases its energy in a fast, hot, violent event; ammonia in a slow, cool, reluctant one; ethanol cools the charge as it evaporates before burning quickly and resisting knock. These are not differences of degree in a single "heat release" quantity: they are qualitatively different pressure events, and the shape of the pressure event is what sets how much of the fuel's energy becomes work.

If chemistry shapes the pressure event, then chemistry can be scheduled. Consider an aircraft climbing through thinning, cooling air, or an engine moving between full-power knock-limited operation and light-load cruise: the binding constraint on efficiency changes (knock, then burn completeness, then pumping loss) and the chemistry best suited to each constraint may differ. The question this paper answers is quantitative: *where, if anywhere, does dynamically optimizing fuel composition buy real efficiency, and how much?*

The study proceeds by simulation, in four stages of increasing fidelity:

1. a cross-class screening (SI piston, diesel, gas turbine, liquid rocket) to find where composition can matter at all;
2. blend scheduling at stoichiometric operation with *real* gasoline/ethanol kinetics;
3. ambient-adaptive fuel selection across sea level, altitude, hot-day, and cold conditions with six fuels;
4. lean-regime fuel switching, hydrogen's natural habitat, with load-matched mission analysis.

The headline result is deliberately two-sided, because the physics is. At stoichiometric operation, dynamic scheduling is worth essentially nothing. In lean operation, it is worth about an efficiency point: the only regime found where the schedule, rather than the fuel choice, is the prize.

The paper is organized as follows. Section 2 sets out the theoretical framework: cycle efficiency and the shape of heat release, the knock/roughness/misfire constraints, and the corner-solution versus regime-access criterion that structures the whole argument. Section 3 describes the methods: the Cantera kinetics pipeline, the calibrated engine cycle model, the evidence-grading vocabulary used for every reported number, and the scope and limits of the screening model. Section 4 reports the results in the four stages above. Section 5 states the refined thesis, gives the economic perspective, lists what would prove the conclusions wrong, and records limitations and next steps. Section 6 concludes. Appendices document the knock-model calibration and the simulation artifact manifest.

2 Theoretical framework

2.1 Cycle efficiency and the shape of heat release

For the ideal Otto cycle, thermal efficiency depends only on compression ratio r and specific-heat ratio γ (Heywood, 1988),

$$\eta_{\text{Otto}} = 1 - r^{1-\gamma}, \tag{1}$$

rising from 49.9% at $r = 10$ to 52.6% at $r = 12$ for $\gamma = 1.3$. Real engines fall well short of Eq. (1) because heat release is neither instantaneous nor adiabatic, and because the usable compression ratio is capped by knock. Both departures are chemistry-dependent.

Finite-rate heat release is conventionally described by the Wiebe function (Wiebe, 1956) for the burned mass fraction,

$$x_b(\theta) = 1 - \exp\left[-a\left(\frac{\theta - \theta_0}{\Delta\theta}\right)^{m+1}\right], \tag{2}$$

where $\Delta\theta$ is the burn duration and θ_0 the start of combustion. Efficiency rewards short $\Delta\theta$ centered near top-dead-center (approaching the constant-volume ideal), and $\Delta\theta$ is set by the fuel's laminar flame speed S_L and its turbulent enhancement, both strong functions of molecule and mixture (Turns, 2012). A fuel that burns twice as fast halves the departure from the ideal constant-volume event. Peak rate of pressure rise $dP/d\theta$, peak pressure, and peak temperature likewise follow from the chemistry: they are the "different types of pressure" of the thesis, and they matter because they are what the knock, noise, and materials constraints act on.

2.2 Constraints: knock, roughness, misfire

Three constraints bound the usable envelope, and each is a different chemistry:

Knock. End-gas autoignition is predicted by the Livengood–Wu integral (Livengood and Wu, 1955),

$$\int_{t_{\text{IVC}}}^{t_{\text{knock}}} \frac{dt}{\tau(P(t), T(t), \mathbf{x})} = 1, \tag{3}$$

where τ is the ignition delay of the *unburned* mixture of composition $\mathbf{x}$. Delay τ varies by orders of magnitude between fuels and exhibits non-monotonic temperature dependence (the negative-temperature-coefficient, NTC, region) for gasoline-like fuels. Production knock prediction commonly reduces this chemistry to octane-number correlations (Douaud and Eyzat, 1978); the mechanism-resolved delays used here keep the composition dependence those correlations

average over. Composition therefore moves the knock limit, which moves the usable compression ratio, which moves efficiency via Eq. (1). This is the master lever of the entire study.

Roughness. Even absent knock, excessive $dP/d\theta$ violates noise, vibration, and harshness limits. The model imposes $dP/d\theta \leq 8$ bar/deg, representative of production NVH bounds. Fast fuels (hydrogen at $\lambda = 1$) hit this wall before they hit knock.

Misfire / burn completeness. Slow fuels at dilute conditions fail to complete combustion within the available crank window. Operating points with $S_L < 15$ cm/s (at in-cylinder conditions) or 10–90% burn duration $\Delta\theta_{10\text{–}90} > 40°$ are excluded, with sensitivity checks reported in Sec. 4.4.

2.3 The scheduling problem: corner solutions vs. regime access

Let $\mathbf{x}$ be the fuel-blend vector (on the simplex), r the compression ratio, θ_s spark timing, λ the relative air–fuel ratio, and $\mathbf{c}$ the operating condition (load, speed, ambient). Indicated efficiency is

$$\eta_{\text{ind}} = f(\mathbf{x}, r, \theta_s, \lambda; \mathbf{c}), \tag{4}$$

and the dynamic-scheduling question is whether

$$\max_{\mathbf{x}(\mathbf{c})} \eta_{\text{ind}} \; - \; \max_{\mathbf{x}=\text{const}} \eta_{\text{ind}} \tag{5}$$

is meaningfully positive: the gain of a condition-dependent fuel schedule over the best fixed fuel, each at its own optimal hardware and calibration.

Two structural outcomes are possible. If one fuel dominates the others on *every* binding constraint at *every* condition, the condition-dependent optimum in Eq. (5) is a **corner solution**: $\mathbf{x}(\mathbf{c})$ is constant, the gain is zero, and the only prize is choosing that fuel (and its compression ratio) once. If instead the *binding constraint changes* with condition (knock at full load, misfire at lean cruise, pumping loss at light load) and different chemistries win under different constraints, the optimum is an interior or switching solution and the schedule has value. The simulations below determine which case holds, in which regime.

3 Methods

3.1 Chemical kinetics

All kinetics are computed in Cantera (Goodwin et al., 2023) with zero-dimensional homogeneous reactors (ignition delay) and freely-propagating premixed flames (laminar flame speed). Three validated mechanisms are used, each in its domain: the LLNL reduced gasoline-surrogate mechanism of Mehl *et al.* (Mehl et al., 2011) (∼679 species, includes ethanol chemistry) for gasoline/ethanol; GRI-Mech 3.0 (Smith et al., 1999) for H_2/CH_4 and their blends; and the 34-species CRECK ammonia mechanism of Stagni and Dinelli (Stagni and Dinelli, 2025) for ammonia and ammonia/hydrogen. Spot checks against literature flame speeds and delay orderings are performed for every fuel/mechanism pairing; cross-mechanism absolute comparisons are treated as ordering-reliable but gap-approximate.

3.2 Engine cycle model

A single-zone crank-resolved SI model is used: isentropic compression of the unburned charge, Wiebe heat release (Eq. (2)) with $\Delta\theta$ scaled from computed S_L, isentropic expansion, plus a pumping-mean-effective-pressure (PMEP) term at part load so the benefit of unthrottled lean

operation appears. Knock onset follows Eq. (3) with ignition-delay tables from the mechanisms above.

Raw homogeneous-reactor delays are systematically shorter than engine knock delays (hot-spot vs. mean charge, $E_a \approx 150$ kJ/mol amplifies the difference); a single global multiplier TAU_MULT = 8.25 (Assumed, C) is applied, calibrated so that PRF95 gasoline at $CR = 10.5$, full load, is borderline knock at MBT spark: the known behavior of a real pump-gasoline engine. One calibration is used across all fuels and mechanisms throughout.

3.3 Evidence grading

Every reported number carries an evidence tag. *Observed* denotes a quantity measured from a primary source; *Reconstructed* a quantity rebuilt from published data; *Modelled* a quantity computed from the simulation pipeline; *Assumed* a quantity fixed by judgment. Source quality follows in parentheses: A, official primary source; B, reputable secondary source or validated mechanism; C, provisional. Thus "+1.20 efficiency points (Modelled, B)" is a pipeline output resting on validated kinetic mechanisms, not a measurement. Uncertainty is stated in the same sentence as each headline number; where no sampling is involved, the hedge names the dominant model-form approximation instead.

3.4 Scope and limits

The model is intentionally a screening tool, not a predictive digital twin: single-zone (no thermal stratification), no ethanol evaporative charge cooling (conservative for ethanol), no EGR or boost, no exhaust-gas temperature or NO_x constraints, $\lambda = 1$ except in the lean study. Absolute efficiency numbers carry these approximations; the *rankings and deltas*, the quantities the thesis turns on, are insensitive to these approximations, and sensitivity analyses bound them. Retrospective analysis: the simulations were completed before this paper was written, and no analysis plan was pre-registered.

4 Results

4.1 Cross-class screening: where can composition matter?

Before detailed modeling, the four engine classes were screened for whether fuel-composition scheduling can buy efficiency *in principle*:

Diesel combustion is mixing-controlled: composition swings ignition delay (∼2× (Modelled, B), ≈9 crank-angle degrees of authority) but brake thermal efficiency is set by the mixing field, not the chemistry. Gas turbines run continuous lean combustion with no knock constraint; a hydrogen blend swings lean flame speed 20.3 → 133.6 cm/s (Modelled, B): valuable for operability margins, worthless for cycle efficiency. Liquid rockets are thermochemically anchored: re-optimizing LOX/LH_2 mixture ratio over a 20× chamber-pressure swing gains 1.14 s I_{sp} (0.25%) (Modelled, B). Only the SI piston engine, whose efficiency is gated by the knock-limited compression ratio, offers a composition lever on efficiency. All subsequent work is SI.

4.2 Blend scheduling at stoichiometry: the corner solution

Using the LLNL gasoline-surrogate mechanism (PRF95 + ethanol blends, ignition-delay grids 750–1400 K, 15–120 atm; sanity check PRF95 at 30 atm/900 K: $\tau = 6.34$ ms (Modelled, B)), each blend was run to its own knock-limited optimum of compression ratio and spark at full load, $\lambda = 1$.

Headline: the dynamic optimum reaches 37.1% indicated at CR 14.5 versus 35.9% for a realistic PRF95 engine at CR 10.5: +1.20 **efficiency points (+3.3% relative)** (Modelled, B;

Table 1: Simulated value of dynamic fuel-composition optimization by engine class.

Engine class	Simulated efficiency gain	Lever
SI piston (dual-fuel)	**+0.95–1.44 pts** (Modelled, B; lean switching, §4.4)	Knock limit → usable CR; lean-regime access
CI diesel	∼0 (BTE) (Modelled, B)	Premixed fraction → noise, NO_x, cold-start only
Gas turbine	∼0 (efficiency) (Modelled, B)	Operability only (blowout and flashback margins)
Liquid rocket	+0.25% I_{sp} (Modelled, B)	None; optimum thermochemically anchored

Notes: Gains are indicated-efficiency points from the screening model; diesel, turbine, and rocket rows are order-of-magnitude screening deltas, read as zero for efficiency purposes.

Source: Kilambi Research simulation archive (`simulations/`).

the comparison engine is a single calibrated reference point, so the second decimal is indicative). But the schedule itself is worth nothing: the best fixed fuel (pure ethanol) already reaches 37.1% at CR 14.0, and the full per-load dynamic schedule ((0.50, 0.50, 0.00, 1.00) ethanol mole fractions across the four load points at CR 14.5) gains only

$$\text{dynamic schedule} - \text{best fixed fuel} = +0.015 \text{ pts} \quad \text{(Modelled, B; below model noise, read as zero)} \tag{6}$$

(+0.04% relative), indistinguishable from noise. At every CR ≤ 12 the dynamic edge is exactly zero and the optimizer does pick ≈ pure ethanol everywhere; the deviations at the CR 14.5 optimum, including a pure-gasoline point at the 0.75-atm load from the high-temperature NTC reversal (ethanol is *more* reactive than PRF95 above ∼1300 K), buy essentially nothing. This is the corner solution of Sec. 2.3: no tradeoff exists to schedule. The +1.20 points are a *fuel-choice plus compression-ratio* prize, won once at the design stage; the dual-fuel hardware's residual value is fuel flexibility (cost, availability), not dynamic optimization.

4.3 Ambient-adaptive fuel selection: the mechanism is real, the value is zero

The thesis was then tested in its most literal form: six fuels (PRF95, ethanol, H_2, CH_4, NH_3, NH_3/H_2 70/30) × four ambients (sea level; 3000 m altitude; hot day; cold), each at its own knock- and roughness-limited optimum. Computed stoichiometric flame speeds confirm the "different types of pressure" premise quantitatively: H_2 234 cm/s with a 21.4 bar/deg, 3° burn; NH_3 7.6 cm/s with a 0.9 bar/deg, 89.5° burn; PRF95/ethanol/CH_4 in between (Fig. 1) (all Modelled, B).

The winner *does* switch with ambient, for exactly the predicted reason: at sea level the binding constraint is knock and the most knock-resistant fuel (CH_4, 37.7% (Modelled, B)) wins; at altitude knock relaxes (PRF95's knock-limited CR moves 10 → 13) and the faster-burning fuel (ethanol, 37.9% (Modelled, B)) wins. Yet ambient-adaptive selection beats the best single fixed fuel (CH_4 at CR 14) by only

$$+0.05 \text{ efficiency points } (+0.14\% \text{ relative}) \quad \text{(Modelled, B)} \tag{7}$$

a margin within the mechanism cross-comparison uncertainty, because the top SI fuels sit within half a point of each other at every ambient (Fig. 2). Hydrogen is infeasible at $\lambda = 1$ everywhere: its wall is roughness, not knock. Ammonia collapses in the cold (28.0% (Modelled, B)). The complementary NH_3/H_2 70/30 pair is the stable compromise, holding 36.6–36.9% across all ambients: hydrogen fixes ammonia's ignition, ammonia tames hydrogen's violence.

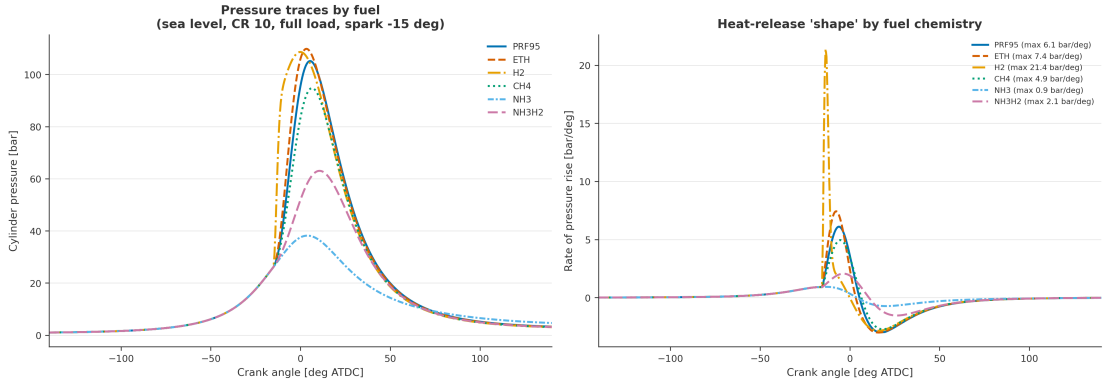


Figure 1: Cylinder pressure traces by fuel at each fuel's knock-limited optimum (sea level, full load). Chemistry visibly changes the shape of the pressure event. But the extreme shapes disqualify their fuels (H_2 too violent at $\lambda = 1$, NH_3 too slow) rather than rewarding switching.

4.4 Lean-regime switching: the schedule finally pays

All prior work ran at $\lambda = 1$, which excluded hydrogen via the roughness cap. Lean operation is hydrogen's natural habitat, and the constraint structure changes there: knock largely disappears, misfire and burn duration become binding, γ rises with leanness, and part-load pumping loss can be avoided by unthrottled lean operation. Five fuels (H_2, CH_4, NH_3/H_2 70/30, H_2/CH_4 50/50, ethanol) were swept over $\lambda = 1.0$–2.2 (plus a hydrogen probe at 2.6) at full and part load, with $\gamma(\phi) = 1.40 - 0.09\phi$ (Assumed, C) and the misfire rule of Sec. 2.2. Flame speeds were computed directly at IVC-like conditions (350 K, 1 atm): e.g. H_2 299 $\rightarrow$ 38 cm/s from $\lambda = 1.0$ to 2.2; CH_4 49 $\rightarrow$ 4.7 cm/s; all match literature (Modelled, B; spot-checked against published flame speeds, Observed, B).

The winner now switches with λ *and* load, and hydrogen is the only fuel that survives lean; every other fuel hits the misfire wall past $\lambda \approx 1.4$. Because the screening matrix conflated λ with load, the honest comparison fixes *load* as the requirement and optimizes fuel, λ, throttle, CR, and spark per fuel (three mission points, ∼41,000 cycle evaluations):

Table 2: Load-matched mission analysis. Load is the requirement; one engine at CR 14, with fuel/λ/throttle/spark optimized per fuel.

Mission point	Winner	Runner-up	Gap (pts)
Full power, 3000 rpm	H_2/CH_4 38.2% (λ=1.0)	CH_4 38.1%	+0.10
Cruise 3 bar, 2000 rpm	**H_2 34.7%** (λ=1.8)	H_2/CH_4 33.0% (λ=1.4)	+1.75
Mid-load 6 bar, 2000 rpm	**H_2 38.5%** (λ=2.2)	H_2/CH_4 37.4% (λ=1.4)	+1.10

Notes: Efficiencies are indicated values from the screening model, rounded to three significant figures. Gaps are adaptive schedule minus best fixed fuel at that mission point.

Source: Kilambi Research simulation archive (`sim4d_mission.csv`); all values Modelled, B.

Adaptive (H_2/CH_4 blend at full power, H_2 lean at cruise/mid-load, one engine at CR 14) versus the best single fixed fuel (H_2/CH_4 at CR 14) gives

$$\boxed{+0.95 \text{ points equal-weighted, } +1.44 \text{ points drive-weighted (5/60/35)}} \quad \text{(Modelled, B)} \quad (8)$$

That is 60× the blend-scheduling value and 30× the ambient-switching value: the first positive scheduling result of the series. At full power the adaptive winner coincides with the best fixed fuel, so the entire gain comes from cruise and mid-load. It is not knife-edged: relaxing the misfire

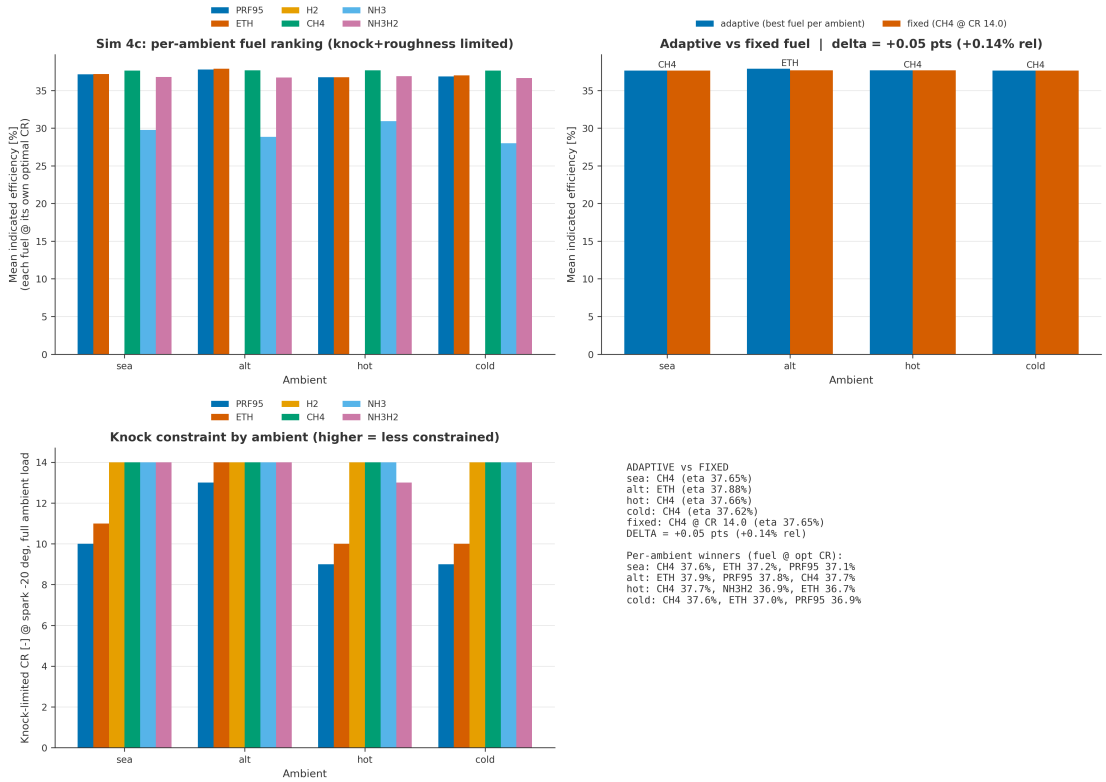


Figure 2: Efficiency ranking by fuel and ambient, each at its own knock/roughness-limited optimum. The winner switches with ambient, but adaptive selection gains only +0.05 points over the best fixed fuel.

rule (40° → 50° burn, 15 → 12 cm/s floor) changes nothing, and halving the lean-γ slope still leaves +0.44/+0.75 points (Modelled, B), so the sign is firm and the magnitude lies between roughly one-half and one-and-one-half points. Lean operation is itself a prize: at matched cruise load, lean H_2 beats the best throttled-stoichiometric option by +1.75 points (Modelled, B), and the best lean point (38.5%) exceeds the best stoichiometric point (38.2%). The fuel switch is what makes this possible: no single fuel covers both full power (H_2 roughness-excluded) and lean cruise (hydrocarbons misfire-walled).

5 Discussion

5.1 The refined thesis

The series began with the thesis that different chemistries create different types of pressure and heat, and that scheduling chemistry against conditions should therefore buy efficiency. The simulations confirm the premise and discipline the conclusion:

- The premise is real. Fuels produce visibly, measurably different pressure events (Figs. 1, 4), the winning fuel switches with ambient for the predicted constraint-shifting reason, and ignition delays swing 2–5× with blending (Modelled, B).
- At stoichiometric operation the scheduling value is ≈ zero (+0.015 pts across operating points, +0.05 pts across ambients; both Modelled, B, both read as zero). The optima are corner solutions: one fuel dominates every binding constraint, so there is nothing to schedule. The prize is fuel choice + compression ratio, co-optimized once (+1.20 pts vs. pump gasoline (Modelled, B); +1.89 pts across ambients (Modelled, B)).

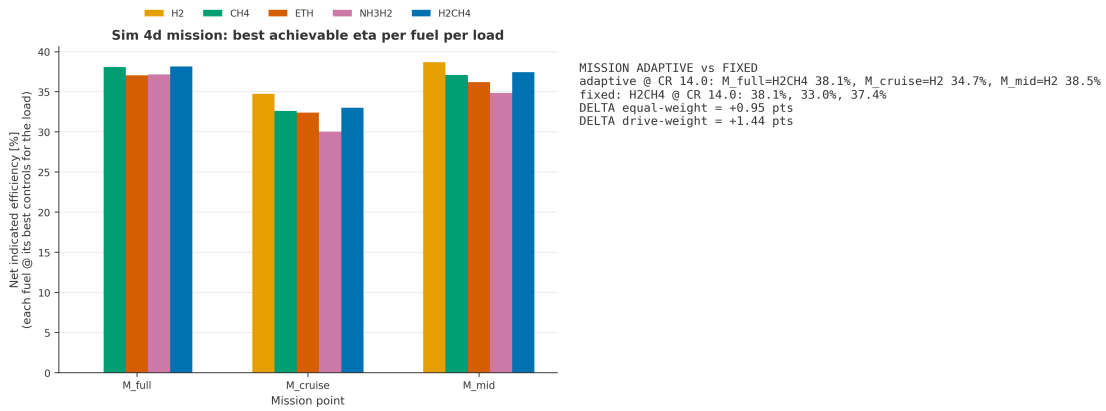


Figure 3: Load-matched mission comparison: adaptive fuel/λ scheduling versus the best single fixed fuel. The cruise and mid-load gains come from hydrogen's exclusive access to the ultra-lean regime.

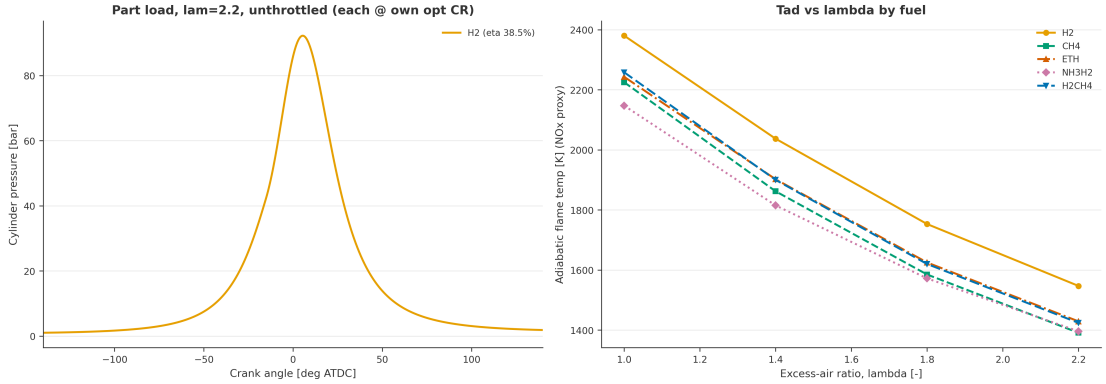


Figure 4: Lean operation at $\lambda = 2.2$, unthrottled: only the hydrogen trace (left) completes combustion in the available window, and adiabatic flame temperature falls with λ for every fuel (right). At $\lambda \approx 2$ only hydrogen-containing fuels remain feasible: regime access, not burn shaping, is the prize.

- In lean operation the answer flips (+0.95/+1.44 pts (Modelled, B)), because the prize was never a better-shaped burn at a fixed point; it is *regime access*. Hydrogen's ultra-lean capability ($\lambda \approx$ 1.8–2.2: fast flame, cool knock-free burn, high γ, minimal throttling) opens a cruise regime at 34.7–38.5% indicated that misfire-walled hydrocarbons cannot enter, while a knock-resistant hydrocarbon blend holds full power where hydrogen is roughness-excluded.

In the language of Sec. 2.3: dynamic fuel-composition optimization pays if and only if the binding constraint changes across the scheduled domain *and* different chemistries win under the different constraints. At $\lambda = 1$ the first condition holds (knock $\rightarrow$ burn completeness across ambient) but the second fails (the top fuels are within half a point under every constraint). In the lean domain both hold. This is offered as a general screening criterion for future composition-control proposals, in any engine class.

5.2 Falsifiability: what would prove this wrong

Three concrete findings would overturn the conclusions above. First, the corner-solution verdict at stoichiometry: a dual-fuel SI engine experiment showing that per-operating-point blend scheduling at $\lambda = 1$ gains more than roughly 0.2 efficiency points over the best fixed fuel at its

own optimal compression ratio, an order of magnitude above the modeled +0.015 (Modelled, B), would falsify it. Second, the knock calibration: if mechanism-resolved ignition delays misorder fuels in the engine environment (for example, if ethanol's high-temperature NTC reversal is a mechanism artifact rather than real chemistry), the CR 14 optimum and the +1.20-point fuel-choice prize move with it. Third, the lean-regime result: if NO_x-constrained optimization or measured lean-limit data showed hydrogen's $\lambda \approx 2$ cruise advantage collapsing (for example, the hydrocarbon misfire wall sitting substantially leaner than modeled, or lean hydrogen's NO_x forcing re-throttling), the +0.95/+1.44-point scheduling gain would shrink toward the stoichiometric verdict. The screening criterion itself is falsified by any engine class in which scheduling pays without a change in the binding constraint across the scheduled domain.

5.3 Economic perspective

A +1.44-point gain on a 38% base is $\approx$ 4% relative fuel savings (Modelled, B): real, but not a breakthrough, and not purchasable on fuel savings alone against dual-fuel hydrogen hardware (high-pressure storage, injection, safety systems). Green hydrogen currently costs several multiples of natural gas or diesel per unit energy (Observed, C: market price surveys move quickly), so the near-term economic case rests on fuel *flexibility* (burning the cheapest available fuel, the established marine dual-fuel model), on stranded-cheap hydrogen co-located with curtailed renewables, or on carbon pricing and compliance markets where the efficiency gain sweetens a decision made for other reasons. For aviation specifically there is no case on efficiency grounds: airliners fly gas turbines, where composition buys $\sim$ zero (Table 1), and hydrogen aviation's rationale is decarbonization, not cycle efficiency.

5.4 Limitations and next steps

This is a screening study, and its boundaries are stated plainly: single-zone modeling; one global knock calibration across three mechanisms (ordering trusted, exact gaps approximate); no charge cooling, EGR, boost, exhaust-temperature, or NO_x constraints; the lean-γ slope moves the 4d magnitude within the bounded range reported. The natural next steps are crank-resolved multi-zone modeling with spray/evaporation physics, a NO_x-constrained lean optimization (lean hydrogen's emissions behavior is its own question), and the decisive test, a dual-fuel SI research engine (hydrogen + methane/hydrogen blend) running λ sweeps to 2.0+ at part load on a dynamometer, with in-cylinder pressure indicating to close the loop between predicted and measured regime access.

6 Conclusion

Fuel chemistry is an engine control variable, but a highly selective one. Across four engine classes, only the spark-ignition piston engine rewards composition scheduling with efficiency. Within SI, scheduling at stoichiometric operation is worthless: the optimum is a corner solution and the prize is fuel + compression ratio chosen once, and ambient-adaptive switching is worth +0.05 points (Modelled, B; read as zero). The exception, and the answer to the question that motivated this work, is lean operation: hydrogen's exclusive access to the ultra-lean regime makes load-adaptive fuel switching worth +0.95 efficiency points (+1.44 drive-weighted) over the best fixed fuel (Modelled, B; magnitude bounded between roughly one-half and one-and-one-half points by the sensitivity variants). Different chemistries do create different types of pressure; their economic and thermodynamic value lies less in reshaping any single combustion event than in reaching operating regimes that a single chemistry cannot reach.

Acknowledgments

The simulations behind this paper were run through Kilambi Research's quantitative research pipeline (Cantera chemical kinetics with the LLNL, GRI-Mech 3.0, and CRECK mechanisms, coupled to a calibrated knock-limited engine cycle model), with AI assistance in code development, analysis, and drafting. All computed results were verified against the source simulation archive, and the figures reproduce directly from the archived run outputs. No external funding was used.

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A Knock-model calibration

Homogeneous-reactor ignition delays underpredict engine knock delays because knock initiates in hot spots, not at the mean charge state. With an effective activation energy $E_a \approx 150$ kJ/mol, the hot-spot/mean offset amplifies to roughly an order of magnitude in τ. The single multiplier TAU_MULT = 8.25 (Assumed, C) applied to all computed delays was fixed by requiring PRF95 at CR 10.5, full load, $\lambda = 1$ to be borderline-knocking at MBT spark: the documented behavior of production pump-gasoline engines. All fuels, blends, mechanisms, and ambients use this one calibration; no per-case tuning was performed.

B Simulation artifact manifest

All primary data behind this paper are retained in the simulation archive's `simulations/` directory:

- `RESULTS.md` (full cumulative record);
- `sim4b_knock.py`, `sim4b_delays.csv`, `sim4b_knock.png`;
- `sim4c_ranking.csv`, `sim4c_shape.csv`, `sim4c_summary.csv`, `sim4c_knock.png`, `sim4c_ptraces.png`;
- `sim4d_knock.py` and `sim4d_mission.py` (plus sensitivity variants); outputs `sim4d_ranking.csv`, `sim4d_mission.csv`, and the `sim4d_*.png` figures;
- mechanisms under `mech/` (LLNL 679-species gasoline, GRI-Mech 3.0, CRECK ammonia).